

ROBERT VOKÁČ, BSc

Czech Republic | +420 704 780 275 | robertvokac@robertvokac.com
github.com/robertvokac | github.com/openeggbert

PROFILE

Software Engineer focused on modern C++, framework design, rendering architecture, and system-level development.

9 years of professional software development experience, primarily in backend systems (Java, SQL). Currently focused on modern C++ through public framework and library projects.

TECHNICAL SKILLS

Languages

C++ (modern C++20/23), SQL, Java

Systems & Tools

CMake, Git, CLion, Bash, Linux, GCC/Clang, SDL 3, SQLite

Core Concepts

- Framework design & reusable abstractions
- Modular architecture & API design
- Rendering architecture
- Plugin systems & extensibility
- Event-driven systems
- Data modeling & practical persistence
- Maintainable code & explicit design

Operating Systems

Linux (daily use), Windows

KEY PROJECT HIGHLIGHTS

- Designed and implemented modular C++ systems from scratch
- Built reusable framework-style abstractions and public C++ libraries
- Implemented plugin-based extensibility and event-driven architecture
- Worked on rendering architecture, API design, and maintainable system design
- Built reusable framework-style abstractions and supporting C++ libraries

PROJECTS

CNA - XNA 4.0 style C++ reimplementations over SDL 3

GitHub: github.com/openeggbert/cna

Long-term public modern C++ project focused on framework design, rendering architecture, and reusable engine-style library design.

Key features:

- C++ reimplementations of an XNA 4.0 style API
- Built over SDL 3 with clear abstraction boundaries between public API and backend implementation
- Practical work in rendering architecture and framework-level design

Mind Net - Modular C++ Platform

GitHub: github.com/robertvokac/mind-net

Designed and implemented a modular C++ platform from scratch.

Key features:

- ~40k lines of modern C++ code
- Multi-module architecture
- Plugin-based extensibility
- Custom data modeling layer defined directly in C++
- Automatic REST API and database layer generation
- Event-driven system with before/after data change triggers
- Dynamic UI generation driven by application metadata
- Role-based authentication and authorization
- Caching mechanisms for performance optimization
- Custom database migration tool and built-in task scheduler
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easy-gl - Graphics support library

GitHub: github.com/openeggbert/easy-gl

- Smaller graphics-focused C++ library used for practical rendering learning and reusable rendering support

cpp-dotnet - .NET-inspired C++ foundation layer

GitHub: github.com/openeggbert/cpp-dotnet

- Partial reimplementations of selected .NET concepts such as exceptions, events, delegates, and basic system types

PROFESSIONAL EXPERIENCE

Software Engineer

Assist, spol. s r. o. — 2017–present

- Developed and maintained production software systems
- Designed and optimized SQL database structures
- Contributed to system architecture and long-term maintainability
- Worked in a team environment using Git and Agile workflows

Technologies: Java, SQL, Git, Bash

EDUCATION

Czech University of Life Sciences Prague

BSc in Computer Science (2014–2017)

Secondary Technical School Hořovice

Informatics in Economics (2008–2012)

LANGUAGES

English – B2 (technical documentation, team communication)